Final Report

The Art of Scientific Computing: Predicting Price Spikes in the National Electricity Market

Author
Victor Li

Subject Handbook code:
COMP-90072

Faculty of Science
The University of Melbourne

Original Version: 1st March 2018

Subject co-ordinators: A/Prof. Suzie Sheehy & Prof. Christian Reichardt

Contents

0.1 Introduction 1
0.2 Data 1
0.3 Exploratory Analysis of Spikes 2
0.4 Hawkes Process 2
0.5 Forecasting Models 2
.1 Boosting Trees 2

Bibliography **2**

0.1 Introduction

Australian energy prices are unremarkable, both in our minds and in scale. Reserved for the eyes of energy retailers, the wholesale price averaged less than $100 in all states that the National Energy Market serviced in the last quarter. [Reg26] Within short time frames, the regional price can jump by one to two orders of magnitude, before it is hard capped by the market regulator. Such extrema are the driving force behind the market's economic actors - retailers who seek to insure themselves against paying such high prices, and generators who will vary generation on *price spike* forecasts. The nature of this report is to answer such: *Given everything observable at the end of a dispatch interval, how well can we predict that the price will be above $300/MWh some moments later?*

0.1.1 The National Electricity Market

0.1.2 Price Spikes

Why do we define the threshold to be $300/MWh? In some ways it is arbitrary - it is no longer the administered price cap [Com26a], and it doesn't compare to the most extreme prices that are reached. However, it was chosen because the threshold is crossed often enough, and also because it is the price level of the standard cap contracts - the call option retail energy providers purchase to insure against risk. [Com26b]

0.1.3 Aim

0.2 Data

AEMO publishes the outcomes of certain variables every 5 minutes. Such quantised data is a dispatch run in its Market Management System (MMS) database. We use seven MMS tables for the year 13 September 2025 00:05 to 13 September 2026 00:00 (105,120 five-minute intervals). All are indexed by SETTLEMENTDATE, the end of the interval (AEST). They differ in data types, and in how many rows they carry per interval:

- one row per region: DISPATCHPRICE (RRP, FCAS prices, the 7-day cumulative price and the administered-pricing flag) and DISPATCHREGIONSUM (demand, available generation, semi-scheduled wind and solar availability, net interchange, battery state)
- one row per interconnector: DISPATCHINTERCONNECTORRES (flow and import/- export limits)
- one row per binding constraint: DISPATCHCONSTRAINT, about 100 million rows over 2,213 distinct constraints
- one row per locally constrained unit or blocked constraint, and one row per interval for the solver summary (CASE_SOLUTION)

Only non-interventioned data (INTERVENTION=0) is used, because that is the run that sets settlement prices, and also because there has not been an intervention in the year interval chosen.

Initial data filtration meant data types with no variance were discarded, as nothing can be gleaned from such. Furthermore, the constraints file reports too many constraints - 100 million rows of data simply isn't feasible in a more complicated model. To resolve this over-reporting, based on the AEMO constraint FAQ [], the associated marginal value of a constraint with regards to the limiting parameters.

0.3 Exploratory Analysis of Spikes

0.4 Hawkes Process

0.5 Forecasting Models

.1 Boosting Trees

Bibliography

[Com26a] Australian Energy Market Commission. *National Electricity Rules*. 2026. URL: https://energy-rules.aemc.gov.au/ner/818 (visited on 10/09/2026).

[Com26b] Australian Energy Market Commission. *Spot and contract markets*. 2026. URL: https://www.aemc.gov.au/energy-system/electricity/electricity-market/spot-and-contract-markets (visited on 10/09/2026).

[Reg26] Australian Energy Regulator. *Industry charts*. https://www.aer.gov.au/industry/wholesale/charts. 2026.

[] URL: https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/system-operations/congestion-information-resource/constraint-faq.